

Algebra 2

UNIT 1: Foundations



ESSENTIAL QUESTION

How can different forms of an equation be used to model real world situations mathematically?

BIG IDEAS

Students recognize different components of an expression and transform linear, literal, and absolute value equations in order to show quantities of interest.

Students construct, interpret, analyze, and solve linear, and absolute value models.

Students understand the connection between matrices and systems of linear equations.

GUIDING QUESTIONS

Content Objectives

- Students write equations to model real-world situations. **A.CED.1., A.CED.2.**
- Students interpret the structure of an expression or equation. **A.SSE.1.**
- Students rewrite an expression or equation in order to highlight a quantity of interest. (point-slope, slope-intercept, standard form). **A.SSE.2.**
- Students explain and justify each step when solving equations (one variable, literal, and absolute value). **A.REI.1., A.REI.2., A.REI.3., A.REI.3a., A.CED.4.**
- Students tell how many solutions an equation has whether or not there are extraneous solutions. **A.REI.3.**
- Students describe the difference between correlation and causation. **S.ID.2.**
- Students calculate r-value (using technology) and use it to evaluate the fit of a linear regression equation. **S.ID.1.**
- Students will solve linear programming problems. **A.CED.3.**
- Students use a matrix to solve a system of equations. **N.VM.6.**
- Students will add, subtract, and multiply matrices, including multiplying a matrix by a scalar. **N.VM.7., N.VM.8.**

Process

- How can you solve equations (one variable, absolute value, literal) and inequalities (one and two variable) and justify your process? **A.REI.1.**
- In context, what do the slope and y-intercept of a linear equation indicate? **A.SSE.1.**
- What is the process to generate a linear regression equation? **S.ID.1.**
- What does the correlation coefficient tell about the fit of the linear regression? **S.ID.1.**
- What are the processes to add, subtract, and multiply matrices? **N.VM.7., N.VM.8.**

Reflective

- How can I use the structure and form of an equation to model mathematics?

- When is it appropriate to use a linear model to represent data?
- How can I apply systems of equations and inequalities that contain restraints to a real world context?
- When is it advantageous to model a real-world situation graphically?

FOCUS STANDARDS

Standards of Mathematical Practice

MP.2 Reason abstractly and quantitatively.

MP.4 Model with mathematics.

Content Standards

A.REI.1. (all) Explain each step in solving a simple equation as following from the equality of numbers asserted at the previous step, starting from the assumption that the original equation has a solution. Construct a viable argument to justify a solution method.

A.REI.2. (all) Apply and extend previous understanding to solve equations, inequalities, and compound inequalities in one variable, including literal equations and inequalities.

A.REI.3. Solve equations in one variable and give examples showing how extraneous solutions may arise.

- **A.REI.3a. (9/10/11)** Solve rational, absolute value and square root equations.

A.SSE.1. (all) Interpret expressions that represent a quantity in terms of its context.

- **A.SSE.1a. (all)** Interpret parts of an expression, such as terms, factors, and coefficients.
- **A.SSE.1b. (all)** Interpret complicated expressions by viewing one or more of their parts as a single entity. For example, interpret $P(1 + r)^n$ as the product of P and $(1 + r)^n$.

A.SSE.2. (all) Use the structure of an expression to identify ways to rewrite it.

A.CED.1. (all) Apply and extend previous understanding to create equations and inequalities in one variable and use them to solve problems.

A.CED.2. (all) Apply and extend previous understanding to create equations in two or more variables to represent relationships between quantities; graph equations on coordinate axes with labels and scales.

A.CED.3. (all) Represent constraints by equations or inequalities, and by systems of equations and/or inequalities, and interpret solutions as viable or non-viable options in a modeling context. For example, represent inequalities describing nutritional and cost constraints on combinations of different foods.

A.CED.4. (all) Rearrange formulas to highlight a quantity of interest, using the same reasoning as in solving equations. For example, rearrange Ohm's law $V = IR$ to highlight resistance R .

S.ID.1. (11) Compute (using technology) and interpret the correlation coefficient of a linear fit.

S.ID.2. (11) Distinguish between correlation and causation.

N.VM.6. (11) Use matrices to represent and manipulate data, (e.g. representing information in a linear programming problem as a matrix or rewriting a system of equations as a matrix.)

N.VM.7. (11) Multiply matrices by scalars to produce new matrices, (e.g. as when all of the payoffs in a game are doubled.)

N.VM.8. (11) Add, subtract, and multiply matrices of appropriate dimensions; find determinants of 2×2 matrices.

UNIT 2: Functions

ESSENTIAL QUESTION

BIG IDEAS

How do we describe a relationship between two variables?

Students describe the relationship between two variables exhibited through a table, graph, equation, and applications.

Students describe features of functions (domain, range, even, odd, transformations).

Students write, evaluate, compose, and transform functions.

GUIDING QUESTIONS

Content Objectives

- Students define the domain and range of a relationship. **F.IF.1.**
- Students use and analyze function notation. **F.IF.2.**
- Students write functions given a set of ordered pairs or a pattern. **F.IF.3.**
- Students will compose functions. **F.BF.1c.**
- Students transform functions, $f(x)$: $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$. *For example, given the graph of a function and a graph of the same function transformed with a vertical shrink by a factor of $\frac{1}{2}$ and a horizontal shift left 4 units, students will describe the transformed function as $\frac{1}{2}f(x+4)$.* **F.BF.3.**
- Students will explore the effects of transformations using technology. **F.BF.3.**
- Students will identify even and odd functions. **F.BF.3.**

Process

- How can you write a function to describe the transformations of a graph? **F.BF.3.**
- What does the graph of a transformed function look like? **F.BF.3.**
- How can you determine a function rule given a set of ordered pairs? **F.IF.3.**
- How do you compose functions? **F.BF.1c.**
- How can you use function notation to interpret a problem situation? **F.IF.2.**
- How do you determine the domain and range of a relationship? **F.IF.1.**

Reflective

- How can I use technology to explore the effects of transformations?
- Can I articulate the difference between an even and odd function?

FOCUS STANDARDS

Standards of Mathematical Practice

MP.1 Make sense of problems and persevere in solving them.

MP.5 Use appropriate tools strategically.

Content Standards

F.IF.1. (all) Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then $f(x)$ denotes the output of f corresponding to the input x . The graph of f is the graph of the equation $y = f(x)$.

F.IF.2. (all) Use function notation, evaluate functions for inputs in their domains, and interpret statements that use function notation in terms of a context.

F.IF.3. (9/10/11) Recognize patterns in order to write functions whose domain is a subset of the integers. (9/10) Limited to linear and quadratic. For example, find the function given $\{(-1, 4), (0, 7), (1, 10), (2, 13)\}$

F.BF.1c. (11) Compose functions. For example, if $T(y)$ is the temperature in the atmosphere as a function of height, and $h(t)$ is the height of a weather balloon as a function of time, then $T(h(t))$ is the temperature at the location of the weather balloon as a function of time.

F.BF.3. (9/10/11) Transform parent functions ($f(x)$) by replacing $f(x)$ with $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with cases and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions from their graphs and algebraic expressions for them. For (9/10) focus on linear, quadratic, and absolute value functions.

UNIT 3: Exponential and Logarithmic Functions

ESSENTIAL QUESTION

What are the connections between exponential and logarithmic functions?

BIG IDEAS

Students model with exponential and logarithmic functions.
Students will understand the relationship between exponential and logarithmic functions.
Students will rewrite exponential and logarithmic functions to solve problems.

GUIDING QUESTIONS

Content Objectives

- Students use transformations on exponential and logarithmic parent functions such as $f(x)$ by replacing $f(x)$ with $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$ for specific values of k (both positive and negative) and also find the value of k given the graphs. **F.BF.3.**
- Students investigate transformations and illustrate an explanation of the effects on the graph using technology. **F.BF.3.**
- Students investigate even and odd functions as it relates to exponential and logarithmic functions from graphs or algebraic expressions. **F.BF.3.**
- Students understand the inverse relationship between exponents and logarithms and use this relationship to solve problems involving logarithms and exponents. **F.BF.5.**
- Students will be able to distinguish between situations that can be modeled with linear functions and with exponential functions. **F.LQE.1.**
- Students will show that exponential functions grow by equal factors over equal intervals. **F.LQE.1a.**
- Students can recognize situations in which a quantity grows or decays by a constant percent rate per unit interval relative to another. **F.LQE.1c.**
- Students will construct exponential functions, given a graph, a description of a relationship, or two input-output pairs (include reading these from a table). **F.LQE.2.**
- Students are able to use the properties of exponents to transform expressions for exponential functions and interpret expressions for exponential functions. **A.SSE.3c., F.IF.8c.**

- Students will graph logarithmic and exponential functions by hand for simple cases and using technology for more complicated cases, emphasizing the inverse relationship with exponentials and logarithms showing key features including intercepts and end behavior. **F.IF.7., F.IF.7c.**

Process

- How do you transform exponential and logarithmic functions or describe the transformations from a graph? **F.BF.3.**
- How can you use technology to investigate the effects of transformations on graphs of exponential and logarithmic functions? **F.BF.3.**
- What does it mean for an exponential or logarithmic function to be even or odd? **F.BF.3.**
- What is the process for writing an exponential function, given a graph, a description, or two points? **F.LQE.2.**
- How does rewriting an exponential function highlight a quantity of interest? **A.SSE.3c., F.IF.8c.**
- How do you graph exponential and logarithmic functions? **F.IF.7., F.IF.7c.**

Reflective

- Can I use the relationship between exponential and logarithmic functions to solve problems?
- Can I differentiate between situations that can be modeled by a linear function and situations that can be modeled by an exponential function?
- Can I compare the graphs of exponential and logarithmic functions?

FOCUS STANDARDS

Standards of Mathematical Practice

MP.3 Construct viable arguments and critique the reasoning others.

MP.7 Look for and make use of structure.

Content Standards

F.BF.3. (9/10/11) Transform parent functions ($f(x)$) by replacing $f(x)$ with $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with cases and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions from their graphs and algebraic expressions for them. For (9/10) focus on linear, quadratic, and absolute value functions.

F.BF.4. Find inverse functions.

- **F.BF.4a. (11)** Write an expression for the inverse of a function.
- **F.BF.4b. (11)** Read values of an inverse function from a graph or a table, given that the function has an inverse.

F.BF.5. (11) Understand the inverse relationship between exponents and logarithms and use this relationship to solve problems involving logarithms and exponents.

F.LQE.1. Distinguish between situations that can be modeled with linear functions and with exponential functions.

- **F.LQE.1a. (11)** Prove that linear functions grow by equal differences over equal intervals, and that exponential functions grow by equal factors over equal intervals.
- **F.LQE.1b. (11)** Recognize situations in which one quantity changes at a constant rate per unit interval relative to another.
- **F.LQE.1c. (11)** Recognize situations in which a quantity grows or decays by a constant percent rate per unit interval relative to another.

F.LQE.2. (11) Construct exponential functions, given a graph, a description of a relationship, or two input-output pairs (include reading these from a table).

A.SSE.3c. (11) Use the properties of exponents to transform expressions for exponential functions. For example, the expression 1.15^t can be written as $(1.15^{\frac{1}{12}})^{12t} \approx 1.012^{12t}$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%.

F.IF.7. Graph functions expressed symbolically and show key features of the graph, by hand in simple cases and using technology for more complicated cases.

F.IF.7b. (11) Graph square root, cube root, and exponential functions.

F.IF.7c. (11) Graph logarithmic functions, emphasizing the inverse relationship with exponentials and showing intercepts and end behavior.

F.IF.8c. (11) Use the properties of exponents to interpret expressions for exponential functions. For example, identify percent rate of change in functions such as $y = (1.02)^t$, $y = (0.97)^t$, $y = (1.01)^{12t}$, $y = (1.2)^{\frac{t}{10}}$, and classify them as representing exponential growth and decay.

UNIT 4: Quadratic Functions

ESSENTIAL QUESTION

How do quadratic functions appear in our world?

BIG IDEAS

Students model using quadratic functions.
Students write quadratic functions in different forms.
Students solve quadratic functions using a variety of methods.
Students incorporate complex numbers into their understanding of quadratic functions.

GUIDING QUESTIONS

Content Objectives

- Students identify complex numbers such that $i^2 = -1$, and recognize that every complex number has the form $a + bi$ with a and b real. **N.CN.1.**
- Students use the relation $i^2 = -1$ and the commutative, associative, and distributive properties to add, subtract, and multiply complex numbers. **N.CN.2.**
- Students will be able to find the conjugate of a complex number. **N.CN.3.**
- Students use the structure of an expression to identify ways to rewrite it. *For example, see $x^2 - y^2$ as $(x - y)(x + y)$.* **A.SSE.2.**
- Students will be able to choose and produce an equivalent form of an expression to reveal and explain properties of the quantity represented by the expression. *For example, rewrite from standard to factored form in order to find the x -intercepts.* **A.SSE.3.**

- Students use completing the square in a quadratic expression to reveal the maximum or minimum value of the function it defines. **A.SSE.3b.**
- Students can explain each step in solving simple equations and construct a viable argument to justify the solution. **A.REI.1.**
- Students solve quadratic equations with complex solutions written in the form $a \pm bi$ for real numbers a and b . **A.REI.5b., N.CN.8.**
- Students will use the method of completing the square to transform and solve any quadratic equation in x into an equation of the form $(x - p)^2 = q$ that has the same solutions. **A.REI.5c.**
- Students will write functions in different but equivalent forms to reveal and explain different properties of the functions. **F.IF.8.**
- Students will use the process of factoring and completing the square in a quadratic function to show zeros, extreme values, and symmetry of the graph, and interpret these in terms of a context. **F.IF.8b.**
- Students will use transformations of parent functions ($f(x)$) by replacing $f(x)$ with $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with cases and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions (where applicable) from their graphs and algebraic expressions for them. **F.BF.3.**

Process

- How do operations apply to complex numbers? **N.CN.2.**
- What is the conjugate of a complex number and how can it be used? **N.CN.3.**
- What is the best method for rewriting a given quadratic expression? **A.SSE.2.**
- How does rewriting an expression help highlight a quantity of interest? **A.SSE.3.**
- How do you solve a quadratic equation with complex solutions? **A.REI.5b., N.CN.8.**
- How does completing the square allow you to solve a quadratic equation? **A.REI.5c.**
- How does technology help you discover the effects of transformations of a graph? **F.BF.3.**

Reflective

- Can I explain complex numbers?
- Can I justify each step in solving a quadratic equation?
- Can I rewrite the equation of a quadratic function to determine the characteristics of its graph?
- How can I model a real world situation with a quadratic function?

FOCUS STANDARDS

Standards of Mathematical Practice

MP.2 Reason abstractly and quantitatively.

MP.6 Attend to precision.

Content Standards

N.CN.1. (11) Know there is a complex number i such that $i^2 = -1$, and every complex number has the form $a + bi$ with a and b real.

N.CN.2. (11) Use the relation $i^2 = -1$ and the commutative, associative, and distributive properties to add, subtract, and multiply complex numbers.

N.CN.3. (11) Find the conjugate of a complex number.

N.CN.8. (11) Solve quadratic equations with real coefficients that have complex solutions.

A.SSE.2. (all) Use the structure of an expression to identify ways to rewrite it.

A.SSE.3. Choose and produce an equivalent form of an expression to reveal and explain properties of the quantity represented by the expression.

A.SSE.3a. (9/10) Factor a quadratic expression to reveal the zeros of the function it defines

A.SSE.3b. (11) Complete the square in a quadratic expression to reveal the maximum or minimum value of the function it defines.

A.REI.1. (all) Explain each step in solving a simple equation as following from the equality of numbers asserted at the previous step, starting from the assumption that the original equation has a solution. Construct a viable argument to justify a solution method.

A.REI.5. Solve quadratic equations and inequalities

- **A.REI.5b. (11)** Solve quadratic equations with complex solutions written in the form $a \pm bi$ for real numbers a and b .
- **A.REI.5c. (11)** Use the method of completing the square to transform and solve any quadratic equation in x into an equation of the form $(x - p)^2 = q$ that has the same solutions.

F.IF.8. Write a function in different but equivalent forms to reveal and explain different properties of the function.

F.IF.8b. (11) Use the process of factoring and completing the square in a quadratic function to show zeros, extreme values, and symmetry of the graph, and interpret these in terms of a context.

F.BF.3. (9/10/11) Transform parent functions ($f(x)$) by replacing $f(x)$ with $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with cases and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions from their graphs and algebraic expressions for them. For (9/10) focus on linear, quadratic, and absolute value functions.

UNIT 5: Polynomial Functions

ESSENTIAL QUESTION

How can we expand our knowledge of functions to include

BIG IDEAS

Students will make connections between the equation and graph of a polynomial function.
Students incorporate complex numbers into their understanding of polynomials.

polynomials?

functions.

Students perform algebraic operations with higher order polynomial equations.

GUIDING QUESTIONS

Content Objectives

- Students will be able to use the structure of an expression to identify ways to rewrite it. *For example rewrite from factored into standard form.* **A.SSE.2.**
- Students factor higher degree polynomials; identifying that some polynomials are prime. *For example identify when a standard form polynomial can be factored.* **A.APR.2.**
- Students apply the Remainder Theorem: For a polynomial $p(x)$ and a number c , the remainder on division by $(x - c)$ is $p(c)$, so $p(c)=0$ if and only if $(x - c)$ is a factor of $p(x)$. **A.APR.3.**
- Students generate polynomial identities from a pattern. *For example, difference of squares, perfect square trinomials, (emphasize sum and difference of cubes in grade 11).* **A.APR.4.**
- Students graph polynomial functions, identify zeros when suitable factorizations are available, and show end behavior. **F.IF.7e.**
- Students will use transformations of parent functions ($f(x)$) by replacing $f(x)$ with $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with cases and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions (where applicable) from their graphs and algebraic expressions for them. **F.BF.3.**
- Students will solve polynomial equations in one variable and give examples of problems with extraneous solutions. **A.REI.3., A.REI.3a.**
- Students solve rational equations. **A.REI.3a.**

Process

- What is the best method for rewriting a given polynomial expression in factored form? **A.SSE.2., A.APR.2., A.APR.4.**
- How do you apply the Remainder Theorem? **A.APR.3.**
- How does technology help you discover the effects of transformations of a graph? **F.BF.3.**
- How can you solve a polynomial equation and interpret its solutions? **A.REI.3.**

Reflective

- How can I rewrite a polynomial in standard form?
- Can I determine the characteristics of a polynomial function?
- Can I explain what an extraneous solution is?

FOCUS STANDARDS

Standards of Mathematical Practice

MP.1 Make sense of problems and persevere in solving them.

MP.8 Look for an express regularity in repeated reasoning.

Content Standards

A.SSE.2. (all) Use the structure of an expression to identify ways to rewrite it.

A.APR.2. (11) Factor higher degree polynomials; identifying that some polynomials are prime.

A.APR.3. (11) Know and apply the Remainder Theorem: For a polynomial $p(x)$ and a number c , the remainder on division by $(x - c)$ is $p(c)$, so $p(c)=0$ if and only if $(x - c)$ is a factor of $p(x)$.

A.APR.4. (9/10/11) Generate polynomial identities from a pattern. For example, difference of squares, perfect square trinomials, (emphasize sum and difference of cubes in grade 11).

F.IF.7e. (11) Graph polynomial functions, identifying zeros when suitable factorizations are available, and showing end behavior.

F.BF.3. (9/10/11) Transform parent functions $f(x)$ by replacing $f(x)$ with $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with cases and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions from their graphs and algebraic expressions for them. For (9/10) focus on linear, quadratic, and absolute value functions.

A.REI.3. Solve equations in one variable and give examples showing how extraneous solutions may arise.

UNIT 6: Radical Functions

ESSENTIAL QUESTION

What connections exist between radical functions and other function families?

BIG IDEAS

Students extend their knowledge of roots of real numbers to the features of a radical function.
Students rewrite expressions involving radicals.
Students graph and solve radical equations.

GUIDING QUESTIONS

Content Objectives

- Students explain how the definition of the meaning of rational exponents follows from extending the properties of integer exponents to those values, allowing for a notation for radicals in terms of rational exponents. *For example, we define $5^{\frac{1}{3}}$ to be the cube root of 5 because we want $(5^{\frac{1}{3}})^3 = 5^{\frac{1}{3} \cdot 3} = 5^1$ to hold, so $(5^{\frac{1}{3}})^3$ must equal 5.* **N.RN.2.**

- Students rewrite expressions involving radicals and rational exponents using the properties of exponents. **N.RN.3.**
- Students solve equations in one variable and give examples showing how extraneous solutions may arise. **A.REI.3.**
- Students solve square root equations. **A.REI.3a.**
- Students solve radical and rational exponent equations and inequalities in one variable, and give examples showing how extraneous solutions may arise. **A.REI.4.**
- Students graph square root and cube root functions. **F.IF.7b.**
- Students transform radical functions ($f(x)$) by replacing $f(x)$ with $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with case and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions from their graphs and algebraic expressions for them. **F.BF.3.**

Process

- How can I use a rational exponent to rewrite an expression or equation in radical form and vice versa? **N.RN.2., N.RN.3.**
- How can you solve a radical equation and interpret its solutions? **A.REI.3., A.REI.3a.**
- How can you solve a radical inequality and interpret its solutions? **A.REI.4.**
- How does technology help you discover the effects of transformations of a graph? **F.BF.3.**

Reflective

- Can I explain what an extraneous solution is?
- How can I rewrite a radical equation?
- Can I determine the characteristics of a radical function?

FOCUS STANDARDS

Standards of Mathematical Practice

MP.6 Attend to precision.

MP.7 Look for and make use of structure.

Content Standards

N.RN.2. (11) Explain how the definition of the meaning of rational exponents follows from extending the properties of integer exponents to those values, allowing for a notation for radicals in terms of rational exponents. For example, we define $5^{\frac{1}{3}}$ to be the cube root of 5 because we want $(5^{\frac{1}{3}})^3 = 5^{\frac{1}{3} \cdot 3}$ to hold, so $(5^{\frac{1}{3}})^3$ must equal 5.

N.RN.3. (11) Rewrite expressions involving radicals and rational exponents using the properties of exponents.
A.REI.3. Solve equations in one variable and give examples showing how extraneous solutions may arise.

- **A.REI.3a.** (9/10/11) Solve rational, absolute value and square root equations.

A.REI.4. (11) Solve radical and rational exponent equations and inequalities in one variable, and give examples showing how extraneous solutions may arise.

F.IF.7b. (11) Graph square root, cube root, and exponential functions.

F.BF.3. (9/10/11) Transform parent functions $f(x)$ by replacing $f(x)$ with $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with cases and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions from their graphs and algebraic expressions for them. For (9/10) focus on linear, quadratic, and absolute value functions.

UNIT 7: Modeling

ESSENTIAL QUESTION

How do you use mathematics to model real world situations?

BIG IDEAS

Students will communicate mathematically.

Students will make choices, assumptions, and approximations given mathematical information.

Students will create appropriate models for various situations.

Students will make connections between the characteristics of the function families.

GUIDING QUESTIONS

Content Objectives

- Students use units as a way to understand problems and to guide the solution of multi-step problems; choose and interpret units consistently in formulas; choose and interpret the scale and the origin in graphs and data displays. *For example, choose an appropriate window to allow for analysis of specific characteristics of a graphed function.* **N.Q.1.**
- Students define appropriate quantities for the purpose of descriptive modeling. *For example, the independent variable is t = time and the dependent variable is s = speed.* **N.Q.2.**
- Students choose a level of accuracy appropriate to limitations on measurement when reporting quantities. *For example, measure the area of a floor plan in square feet not square centimeters.* **N.Q.3.**
- Students understand that the graph of an equation in two variables is the set of all its solutions plotted in the coordinate plane, often forming a curve (which could be a line). **A.REI.8.**
- Students solve an equation $f(x) = g(x)$ by graphing $y = f(x)$ and $y = g(x)$ and find the x -value of the intersection point. Include cases where $f(x)$ and/or $g(x)$ are linear, polynomial, rational, absolute value, exponential, and logarithmic.

exponential, and logarithmic functions. **A.REI.9.**

- For a function that models a relationship between two quantities, students interpret key features (expressions, graphs and tables in terms of the quantities, and sketch graphs showing key features given a description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior. **F.IF.4.**
- Students relate the domain of a function to its graph and, where applicable, to the quantitative relationship it describes. *For example, if the function $h(n)$ gives the number of person-hours it takes to assemble n engines in a factory, then the positive integers would be an appropriate domain for the function.* **F.IF.5.**
- Students calculate and interpret the average rate of change of a function (presented symbolically or in a table) over a specified interval. Estimate the rate of change from a graph. **F.IF.6.**
- Students compare properties of two functions using a variety of representations (algebraically, graphically, numerically in tables, or by verbal descriptions). *For example, a quantity increasing exponentially eventually exceeds a quantity increasing linearly.* **F.IF.9.**
- Students use functions to model real-world relationships. **F.BF.1.**
- Students determine an explicit expression, a recursive function, or steps for calculation from a context. **F.BF.1b.**

Process

- What is an appropriate domain and range for a given situation? **N.Q.1.**
- What is the independent quantity of a situation? What is the dependent quantity of a situation? **N.Q.2.**
- What does the graph of an equation represent? **A.REI.8.**
- How do you estimate, calculate, and interpret the rate of change of a function? **F.IF.6.**
- What function family best represents a situation? **F.BF.1.**
- How can a situation be represented algebraically? **F.BF.1b.**

Reflective

- Can I choose units of measure appropriate for a given situation?
- Can I determine intersection points of two functions and interpret its meaning in context?
- Can I determine and interpret key features of a graph?
- Can I compare properties of two functions?

FOCUS STANDARDS

Standards of Mathematical Practice

MP.3 Construct viable arguments and critique the reasoning of others.

MP.4 Model with mathematics.

Content Standards

N.Q.1. (all) Use units as a way to understand problems and to guide the solution of multi-step problems; choose and interpret units consistently in formulas; choose and interpret the scale and the origin in graphs and data displays.

N.Q.2. (all) Define appropriate quantities for the purpose of descriptive modeling.

N.Q.3. (all) Choose a level of accuracy appropriate to limitations on measurement when reporting quantities.

A.REI.8. (all) Understand that the graph of an equation in two variables is the set of all its solutions plotted in the coordinate plane, often forming a curve (which could be a line).

A.REI.9. (9/10/11) Solve an equation $f(x) = g(x)$ by graphing $y = f(x)$ and $y = g(x)$ and find the x-value of the intersection point. Include cases where $f(x)$ and/or $g(x)$ are linear, polynomial, rational, absolute value, exponential, and logarithmic functions.

F.IF.4. (all) For a function that models a relationship between two quantities, interpret key features of expressions, graphs and tables in terms of the quantities, and sketch graphs showing key features given a description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior; and periodicity.

F.IF.5. (all) Relate the domain of a function to its graph and, where applicable, to the quantitative relationship it describes. For example, if the function $h(n)$ gives the number of person-hours it takes to assemble n engines in a factory, then the positive integers would be an appropriate domain for the function.

F.IF.6. (9/10/11) Calculate and interpret the average rate of change of a function (presented symbolically or in a table) over a specified interval. Estimate the rate of change from a graph. (9/10) limited to linear functions.

F.IF.9. (all) Compare properties of two functions using a variety of representations (algebraically, graphically, numerically in tables, or by verbal descriptions). For example, a quantity increasing exponentially eventually exceeds a quantity increasing linearly.

F.BF.1. Use functions to model real-world relationships.

F.BF.1b. (11) Determine an explicit expression, a recursive function, or steps for calculation from a context.

